

From sensor signatures to survival curves: a physics-regularized digital twin for milling tool wear and tool-life forecasting

Amrit Lahari^a, Sukrit Agarwal^a, Pushkar Agrawal^a, Amit Kumar Jain^{a,*}

^aDepartment of Mechanical Engineering, Birla Institute of Technology and Science, Pilani, India

*Corresponding supervisor.

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ABSTRACT

This paper reports the integrated research workflow developed for milling tool wear and tool-life forecasting. The work begins with NASA milling exploratory analysis, feature extraction and leave-one-case-out wear modeling, expands to cross-dataset health baselines, and then connects those empirical signals to physics-regularized lifetime estimation. Three complementary public milling sources are used: NUA/UniWear orthogonal milling records for process-load effects, PHM2010 cutter trajectories for early-regime transfer, and NASA milling run-to-failure data for late-stage acceleration, sensor evidence and material effects. The central objective is not only to minimize pointwise wear error, but to construct an interpretable, simulator-ready model that links process variables, sensor signatures, uncertainty and threshold-crossing life. The proposed framework combines (i) dataset-role analysis, (ii) grouped validation for wear regression and failure classification, (iii) time-domain, FFT and variational-mode-decomposition features, (iv) a physics-regularized piecewise wear law with early exponent $m_e = 0.64$ and late exponent $m_l = 1.29$, (v) bootstrap intervals and residual diagnostics, and (vi) lifetime extraction by training wear models and inverting them into crossing times. Feed emerges as the most reliable local load driver. VMD improves the strongest absolute feature-wear correlation from 0.4806 to 0.5607. The best pointwise wear regressor, Gradient Boosting, does not give the best event-life forecast after inversion, showing that optimizing wear RMSE is not equivalent to optimizing tool-life prediction. The resulting digital-twin prototype is suitable for research simulation, scenario ranking and explanation, but requires larger full-life validation before industrial deployment.

1 Introduction

Milling tool wear is not a single regression problem. A cutting edge can spend a long interval in slow early wear, then move into accelerated damage where a small amount of additional cutting time produces a large increase in flank wear. A realistic monitoring system therefore has to answer four coupled questions. First, it must describe the physical progression from early wear to accelerated wear. Second, it must explain how spindle speed, feed and depth alter the rate of that progression. Third, it must identify which vibration, acoustic-emission or current signatures provide evidence that the tool is moving toward failure. Fourth, it must translate a trained wear model into a practical replacement-time or remaining-useful-life estimate.

The project reported here was built around those questions. The workflow starts with NASA milling exploratory analysis, hand-crafted time and frequency features, leave-one-case-out

wear modeling, feature selection and interpretation. It then expands from one dataset to cross-dataset health baselines for PHM2010, NASA and UniWear, fits a physics-regularized piecewise wear equation, adds uncertainty intervals, calibration checks and residual diagnostics, and finally trains wear models that are inverted into threshold-crossing estimates and compared against accelerated-failure-time equations. This matters because a model that predicts flank wear well at individual samples is not necessarily a model that predicts the crossing time accurately.

The main contributions are as follows. A NASA sensor-modeling benchmark establishes the limits of raw feature regression under case-held-out validation. A dataset-role framework separates early-wear load evidence, early-exponent transfer evidence and late-stage acceleration evidence instead of pooling heterogeneous data blindly. A compact piecewise wear law explains why a single exponent is not enough. A

cross-dataset health pipeline evaluates regression and failure classification under grouped validation. A VMD-based feature pass shows that modal features can improve wear correlation over raw signal descriptors. A lifetime-model extraction pipeline trains multiple wear models, inverts them into crossing estimates and exposes the mismatch between wear RMSE and life accuracy. Finally, bootstrap uncertainty, residual diagnostics and early-window calibration tests describe where the present digital twin is stable and where it remains under-identified.

2 Background and research gap

Tool condition monitoring literature has traditionally emphasized two tasks: estimating the current wear state from sensor signals and classifying a tool as healthy or faulty. Those tasks are important, but they are incomplete for replacement planning. A factory decision is a time-to-event decision: when is the tool likely to cross a failure threshold, and how much uncertainty surrounds that crossing time?

Survival analysis is useful because tool-life data often contain censoring. A cutter may not reach the selected wear threshold before the experiment ends. Treating that record as an ordinary non-failure discards information and biases the fitted life model. Accelerated-failure-time modeling gives a compact way to express how process variables multiply or compress expected lifetime. The project uses this idea directly for NUAA and NASA life equations.

Physics-informed learning is also necessary because public milling datasets are heterogeneous. Some sources contain many process settings but do not run all tools to severe failure. Others contain rich run-to-failure trajectories but fewer process variables. A single black-box regression over all records would conflate early and late wear physics, sensor differences, units and thresholds. The research gap addressed here is therefore not simply “train a better regressor.” The gap is to connect sparse experimental evidence, physics-guided structure, sensor features, uncertainty and threshold-crossing life in a single research-grade workflow.

3 Data and evidence structure

3.1 Dataset roles

Three datasets form the core evidence base, but they are not interchangeable. NUAA/UniWear carries the process variation needed to fit the load law. PHM2010 checks whether the early exponent transfers beyond one source. NASA provides severe wear progression, material effects and multi-sensor evidence. Table 1 summarizes these roles.

The failure thresholds are dataset-specific because the measurement scale and experimental intent differ. These thresholds are operational labels for local model comparison, not universal industrial replacement limits. PHM2010 uses 0.150 mm, NASA uses 0.300 mm and UniWear uses 0.220 mm. The corresponding failure rates are 0.1725, 0.4795 and 0.2743 respectively.

3.2 Feature families

The feature pipeline extracts compact descriptors from time-domain and frequency-domain signals. Time-domain level features include mean, root-mean-square and absolute mean.

Shape features include skewness, kurtosis, crest factor and quantiles. Frequency features include FFT energy, dominant FFT bin, centroid and entropy. Process variables such as speed, feed, depth and material anchor the statistical features in machining physics.

3.3 Variational mode decomposition

The NASA signal pass applies variational mode decomposition to vibration, acoustic emission and spindle-current channels. The settings are $K = 5$, $\alpha = 2000$ and $\tau = 0$. The pass produces 390 modal features over vib_spindle, vib_table, AE_table, AE_spindle, smcAC and smcDC. The strongest raw feature correlation with wear is $|r| = 0.4806$, while the strongest VMD feature reaches $|r| = 0.5607$. This is a 16.7% improvement in the best absolute correlation.

Feature selection and SHAP analysis support the same conclusion: useful predictors are not only raw amplitude measures. Wear information appears in modal center frequency, entropy, kurtosis and acoustic-emission frequency terms. This is important for the simulator because it gives a richer evidence layer than simply measuring one RMS channel.

4 Physics-regularized wear model

4.1 Rationale

The piecewise law is the central modeling choice. A single power-law fit forces early and late wear into one compromise exponent. That is unsuitable for tool-life forecasting because the replacement decision is dominated by behavior near the threshold. The project therefore separates the early sub-linear regime from the accelerated late regime while keeping the equation compact enough for simulator use.

The model uses a reference-point process-load function $\phi(n, f_z, a_p)$ and regime-specific exponents:

$$VB(t) = \begin{cases} k_e \phi(n, f_z, a_p) t^{m_e}, & t \leq t_{tr}, \\ VB_{tr} + k_l (t - t_{tr})^{m_l}, & t > t_{tr}. \end{cases} \quad (1)$$

The transition value is

$$VB_{tr} = k_e \phi(n, f_z, a_p) t_{tr}^{m_e}, \quad (2)$$

and the early condition intensity is

$$\phi = \left(\frac{n}{1800} \right)^{0.359967} \left(\frac{f_z}{0.05} \right)^{3.956013} \left(\frac{a_p}{3.0} \right)^{0.688073}. \quad (3)$$

The estimated early law is

$$VB(t) = 0.018580943 \phi t^{0.64}. \quad (4)$$

4.2 Evidence for the exponents

The exponent values are not fitted from one pooled curve. NUAA is used for the load-dependent early law, PHM2010 checks early-regime transfer and NASA provides late acceleration. The resulting evidence is summarized in Table 5.

The fitted condition law is strongest in feed. This result is physically plausible because feed directly changes chip thickness and cutting load per tooth. Speed and depth also matter, but in the local NUAA design the feed exponent is more stable and much larger than the others. Bootstrap diagnostics later confirm that feed is the most consistently identified process variable in this design.

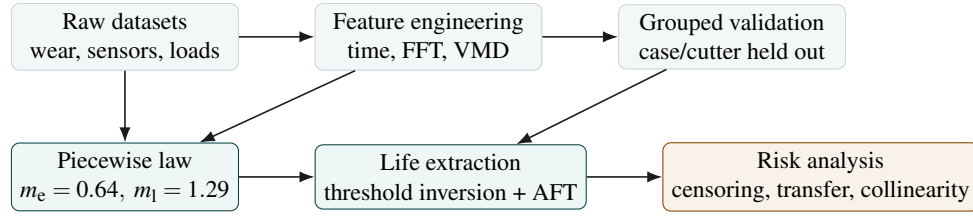


Figure 1: Research architecture followed in the project. The workflow separates signal evidence, physics-regularized equation building, grouped validation, lifetime inversion and risk interpretation.

Table 1: Primary datasets and their research roles.

Dataset	Local role	Main measurements	Why it matters
NUAA / UniWear	Early wear and process-condition law	Wear, spindle speed, feed per tooth, axial depth	Supplies process variation needed to fit the load-intensity function and lifetime surfaces. It includes censored runs, making survival-aware modeling necessary.
PHM2010	Early-regime transfer check	Cutter wear trajectories and derived features	Tests whether the early exponent learned from NUAA is a local artifact or a transferable early-wear slope.
NASA Milling	Late-stage acceleration and material behavior	Vibration, acoustic emission, spindle current, wear, feed, depth, material	Contains severe wear progression and rich sensors, making it the best local source for accelerated wear, frequency-domain evidence and material-aware survival behavior.

Table 2: Dataset-specific health labels.

Dataset	Samples	Threshold	Failure rate
PHM2010	945	0.150 mm	0.1725
NASA	146	0.300 mm	0.4795
UniWear	649	0.220 mm	0.2743
Derived threshold-label counts.			
Dataset	Failed	Non-failed	
PHM2010	163	782	
NASA	70	76	
UniWear	178	471	

Check: $163/945 = 0.1725$, $70/146 = 0.4795$ and $178/649 = 0.2743$. Counts are derived from the sample totals and rounded failure rates above.

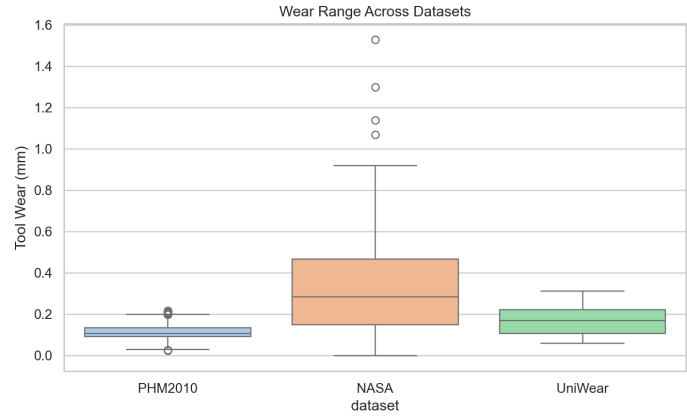
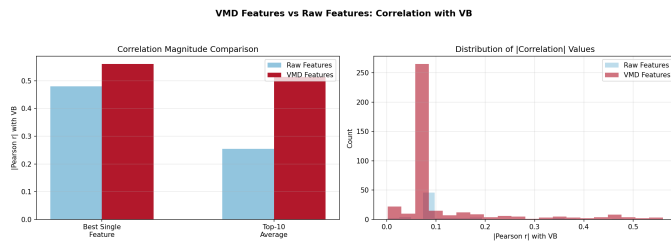


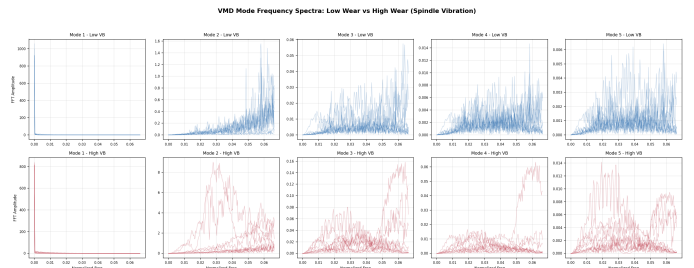
Figure 2: Wear distributions across local datasets. Different ranges motivate dataset-specific thresholds and role-specific modeling.

Table 3: Feature families and interpretation.

Family	Examples	Tool-wear interpretation
Time-domain level	mean, RMS, absolute mean	Captures load and energy growth as cutting becomes less efficient.
Time-domain shape	skewness, kurtosis, crest factor, quantiles	Captures impulsive events, chatter bursts, rubbing and intermittent edge damage.
Frequency-domain energy	FFT energy, peak bin, centroid, entropy	Captures shifts in vibration and acoustic-emission structure as wear changes cutting dynamics.
VMD mode features	mode energy, entropy, center frequency, mode kurtosis	Separates mixed sensor signals into interpretable bands, improving some wear correlations over raw features.
Process variables	speed, feed, axial depth, material	Enable scenario simulation and prevent the model from becoming purely sensor-specific.



(a) Raw versus VMD correlation

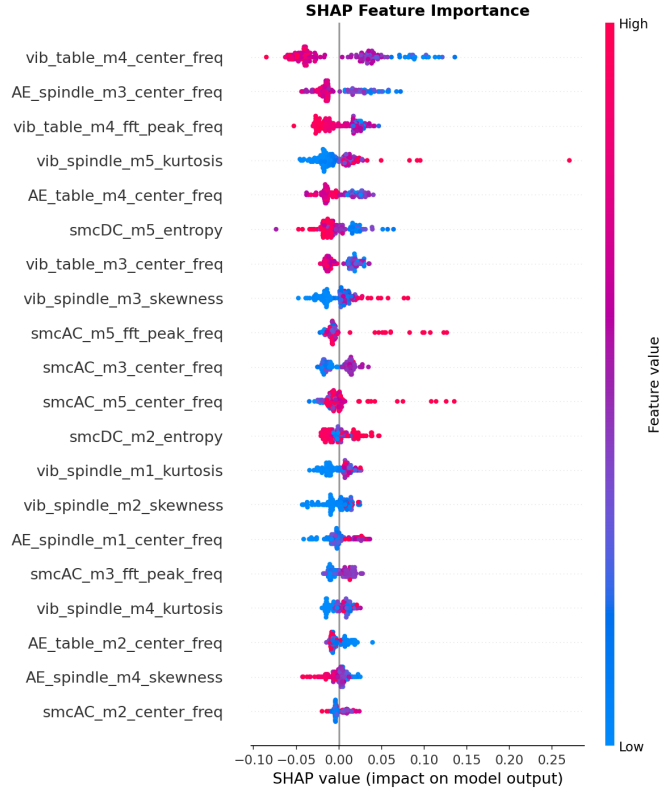


(b) Mode spectra

Figure 3: VMD separates broadband machining signals into mode-specific frequency content before feature extraction.

Table 4: Top VMD features by absolute correlation with NASA flank wear.

Feature	r	$ r $	Interpretation
vib_spindle_m3_skewness	0.5607	0.5607	Wear increases asymmetric spindle vibration bursts.
smcDC_m5_entropy	-0.5604	0.5604	Current structure becomes less evenly distributed in the high-order mode.
vib_spindle_m5_kurtosis	0.5542	0.5542	Impulsive high-frequency vibration grows with tool damage.
vib_table_m4_center_freq	-0.5287	0.5287	Table vibration frequency content shifts as cutting stiffness changes.
vib_spindle_m1_skewness	0.5079	0.5079	Low-mode spindle asymmetry tracks progressive wear.

**Figure 4:** SHAP summary for the NASA modeling pass. High-impact predictors include VMD center-frequency, entropy and kurtosis terms.**Table 5:** Evidence used to select the piecewise wear exponents.

Dataset	Regime	Exp.	RMSE	Role
NUAA	Early	0.6347	0.0211	Main sub-linear early law.
PHM2010	Early	0.6471	0.0136	Transfer check.
NASA	Late	1.2873	0.0438	Accelerated late behavior.

4.3 Material effects

NASA is treated as a separate late-stage and material source. Its feed units, material families, fixed speed and wear scale differ from NUAA, so direct pooling would be unsafe. The NASA late-stage comparison suggests that steel has a stronger late-stage wear factor than cast iron in the local model. This material evidence is used to keep the simulator from treating all late-stage wear as material-independent acceleration.

5 Model validation and results

5.1 NASA sensor-modeling benchmark

The initial single-dataset modeling pass used the NASA milling records to test whether hand-crafted sensor features could predict flank wear under case-held-out validation. The inventory contained 167 runs, of which 146 included VB labels across 16 cases. The observed wear range was 0.000–1.530 mm, with mean 0.338 mm and standard deviation 0.261 mm. Before modal decomposition, the strongest simple raw-feature association with wear was vib_spindle_kurtosis with $r = 0.3823$. This early result motivated the later use of frequency and modal descriptors rather than relying only on raw amplitude summaries.

Table 6: NASA leave-one-case-out wear-regression benchmark using 29 engineered features.

Model	RMSE	MAE	R^2	MAPE
XGBoost	0.1892	0.1500	-0.0451	48.4%
SVR	0.2114	0.1733	-0.1602	61.1%
Random Forest	0.1758	0.1429	0.1824	47.8%
Gradient Boosting	0.1847	0.1460	0.0287	50.1%

The best initial NASA regressor was Random Forest, but the modest R^2 and high percentage error showed that a direct black-box wear regressor was not sufficient for replacement planning. That finding is carried forward into the rest of the work: all later model comparisons use grouped validation, dataset-specific thresholds and explicit checks for whether low wear error transfers into useful life estimates.

5.2 Cross-dataset health baselines

The cross-dataset pipeline evaluates wear regression and failure classification with grouped validation. Grouped validation is essential because adjacent samples from the same cutter or case are not independent deployment tests. The goal is to estimate transfer to unseen experiments rather than interpolation inside one trajectory.

Table 7: Cross-dataset wear regression and failure classification.

Dataset	Samples	Features	RMSE	ROC-AUC
PHM2010	945	105	0.0181	0.9671
NASA	146	94	0.1728	0.9141
UniWear	649	46	0.0475	0.7350

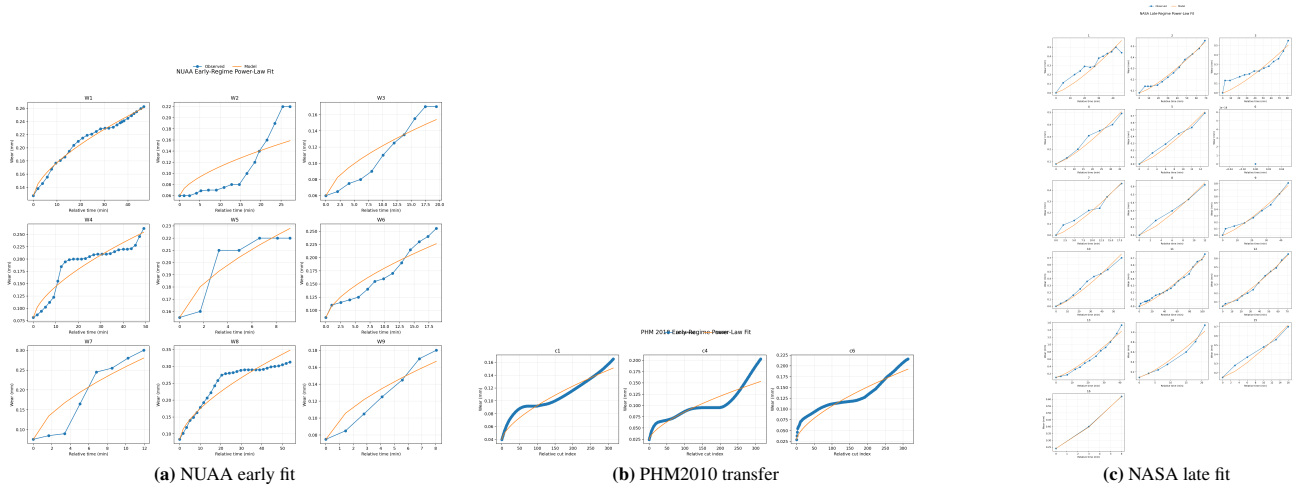
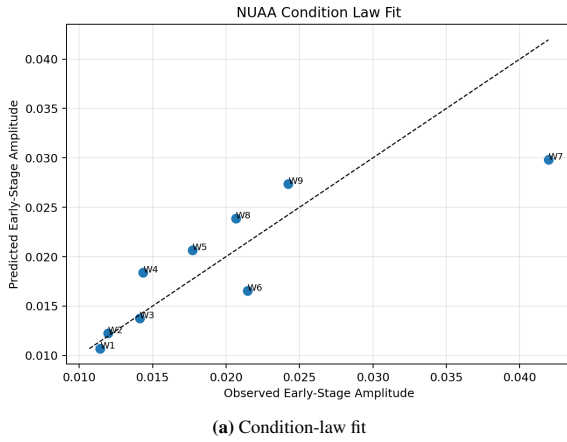
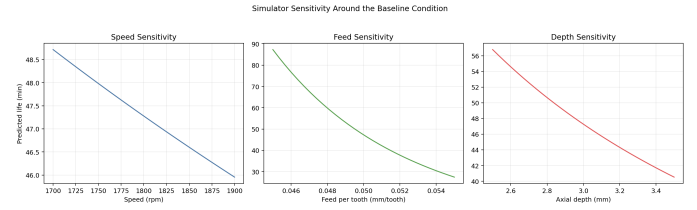


Figure 5: Role-specific fits used to ground the early and late exponents.



(a) Condition-law fit



(b) Life sensitivity curves

Figure 6: Feed dominates the local life sensitivity, while speed and depth remain less stable because of narrower design variation and collinearity.

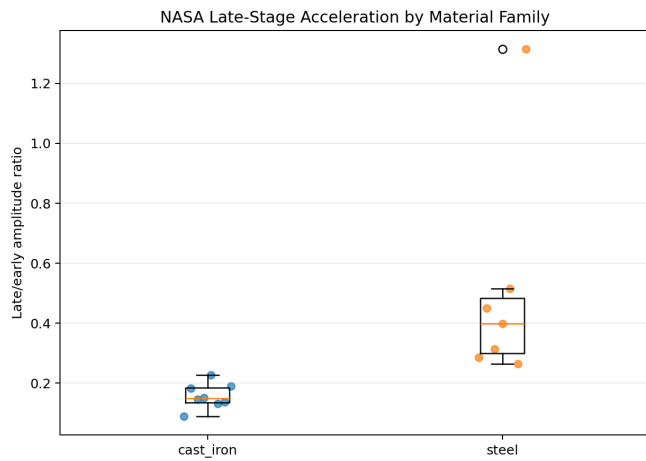


Figure 7: NASA late-stage material behavior. Steel shows a stronger late-stage factor in the local model.

The failure-marker plots show that failure labels are usually preceded by increasing energy, dispersion or impulsiveness, but the exact marker family varies by dataset. This is consistent with machining physics: tool wear changes chip formation, friction, heat generation and dynamic stiffness, and the resulting evidence can appear in vibration, acoustic emission, force/current or frequency features depending on the available sensors.

5.3 Coefficient uncertainty

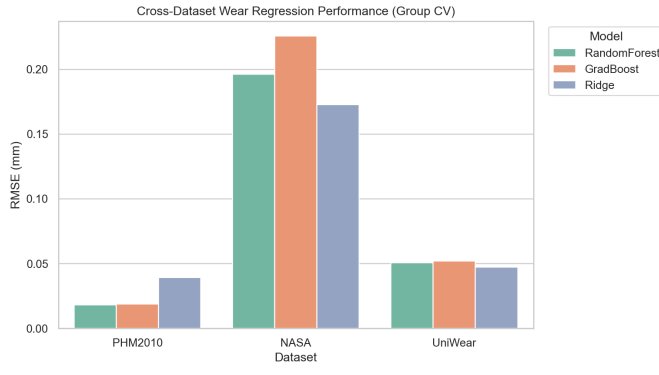
The extended research pass bootstraps the NUAAs condition law with valid-design filtering. This is necessary because small experimental designs can produce rank-deficient or unstable re-samples. The results show a stable feed exponent, while speed and depth remain more uncertain.

Table 8: Bootstrap coefficient summary for the NUAAs condition law.

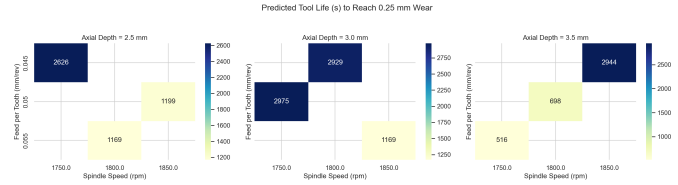
Quantity	2.5%	Median	97.5%
k_{early}	0.01558	0.01783	0.02183
Speed exponent	-7.49808	1.16739	8.04424
Feed exponent	2.04563	3.99243	6.68021
Depth exponent	-0.59012	0.51957	1.92473
Amplitude R^2	—	0.89715	—

5.4 Life intervals and calibration

Bootstrap draws are propagated into low-load, baseline and high-load life scenarios. The baseline median life is about 50.2

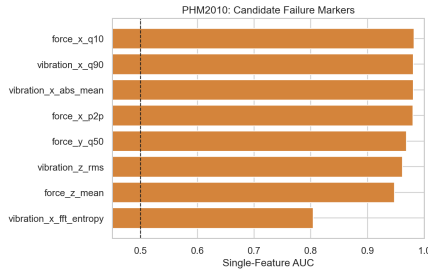


(a) Regression RMSE

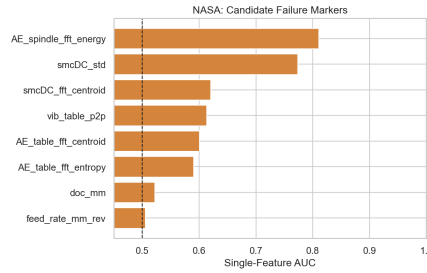


(b) NUA life heatmap

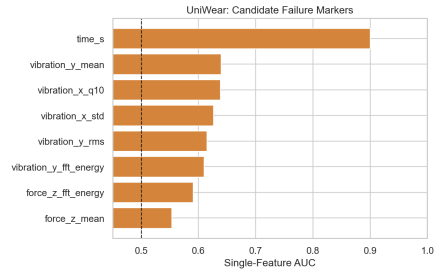
Figure 8: Cross-dataset baselines and process-condition sensitivity. NASA remains the hardest local regression regime because it includes material variation and severe wear.



(a) PHM2010



(b) NASA



(c) UniWear

Figure 9: Failure-marker summaries across datasets. The plots support energy, dispersion, crest and modal frequency indicators as health signals.

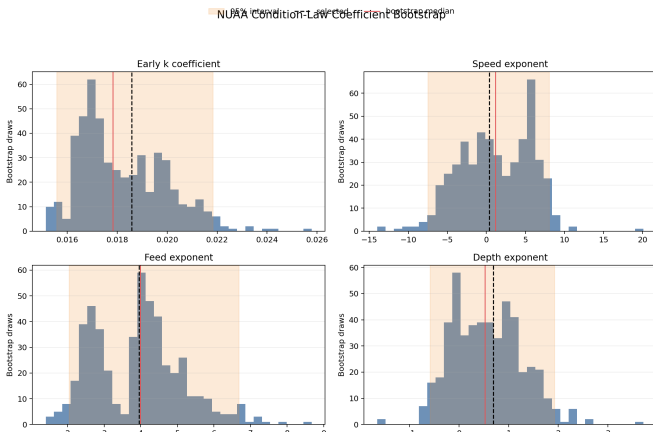


Figure 10: Bootstrap distributions for condition-law coefficients. Feed is the most consistently identified operating variable.

min. The low-load condition extends median life to about 103.6 min, while the high-load condition compresses median life to about 25.0 min.

Table 9: Bootstrap life prediction intervals.

Scenario	Speed	Feed	Depth	2.5%	Median	97.5%
Low load	1750	0.045	2.5	59.797	103.644	247.675
Baseline	1800	0.050	3.0	37.422	50.227	61.177
High load	1850	0.055	3.5	11.514	24.987	38.046

The project also tested whether early observations from each run could calibrate the global equation through a per-run multiplier. The result was negative for the local NUA early regimes. At cutoff fractions of 0.25, 0.40 and 0.60, calibrated RMSE values were 0.05045, 0.04750 and 0.04230, while uncalibrated RMSE values were 0.04047, 0.02508 and 0.02819. This prevents an overly optimistic simulator story: short-window scalar

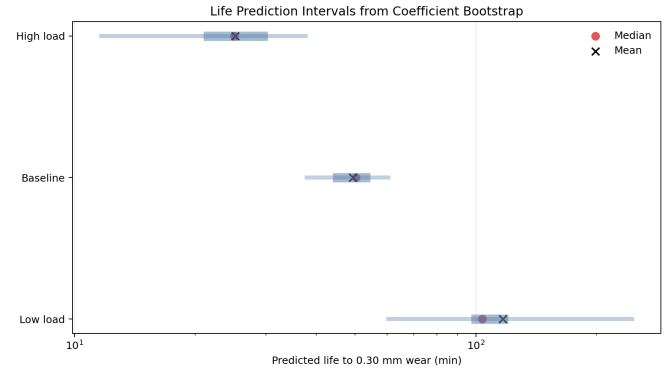


Figure 11: Load-dependent life intervals. The high-load scenario roughly halves baseline median life.

calibration can overcorrect the global model. Future calibration should use monotonic constraints, Bayesian updating or a state-space formulation.

5.5 Residual diagnostics

Residual diagnostics show that the piecewise equation is most accurate in PHM2010 and NUA early regimes, and least accurate in NASA, where severe wear, material variation and late-stage behavior are stronger.

Table 10: Residual diagnostics for the piecewise wear equation.

Dataset	Points	Bias	MAE	RMSE	95% abs.
NASA	146	0.00763	0.03149	0.04376	0.08825
NUAA	152	-0.00162	0.01568	0.02113	0.04087
PHM2010	945	0.00104	0.01070	0.01362	0.02437

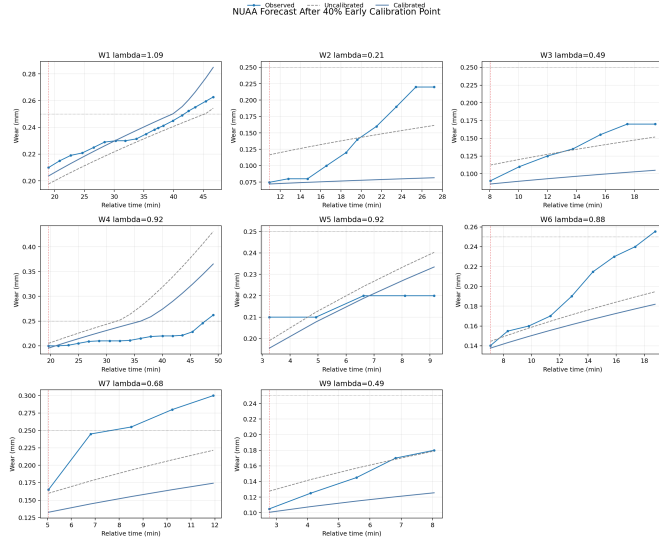


Figure 12: NUA calibration diagnostic. Short-window scalar calibration can worsen holdout forecasts.

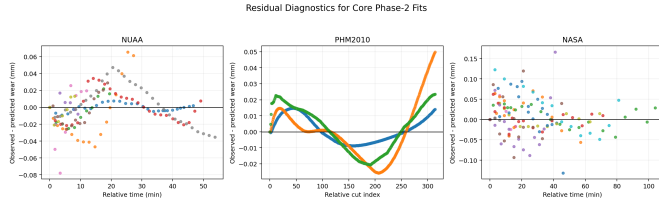


Figure 13: Residual diagnostics across datasets. The NASA tail remains the hardest local regime.

6 Lifetime modeling

6.1 Threshold labels

The lifetime rework directly addresses whether tool life can be estimated by training models and extracting results from those models. It can, but the target changes. Life labels are built by finding when each run crosses a dataset-specific wear threshold. Runs that do not cross before the observation end are treated as censored. NUA and NASA lives are in minutes, while PHM2010 uses cutter cut index because its native time base differs.

Table 11: Threshold-crossing life labels.

Dataset	Records	Threshold	Events	Censored
NUAA	9	0.250 mm	5	4
NASA	15	0.300 mm	15	0
PHM2010	3	0.150 mm	3	0

6.2 Training wear models and inverting life

NUAA wear models are trained under leave-one-experiment-out validation and then inverted to estimate threshold-crossing life. The inversion step exposes a negative result that is important for model selection. Gradient Boosting has the best pointwise wear RMSE, but its event-life MAE is worse than the compact piecewise equation and Ridge. Therefore, the lifetime problem cannot be judged only by VB regression error. Smoothness and bias near the threshold matter.

6.3 NUA accelerated-failure-time equation

The NUA censored accelerated-failure-time model is

$$T_{\min} = 24.93 \left(\frac{n}{1800} \right)^{-0.0187} \left(\frac{f_z}{0.05} \right)^{-1.7662} \left(\frac{a_p}{3.0} \right)^{-0.3768}, \quad (5)$$

with $\sigma = 0.5291$ and negative log likelihood 21.2365. The AFT feed exponent is weaker than the dense condition-law feed exponent because it models sparse threshold-crossing events with censoring, not dense early wear amplitudes.

6.4 ML-inverted and observed life equations

The Ridge response surface is inverted over a grid of operating scenarios and then compressed into a Taylor-like equation:

$$T_{\min} = 47.52 \left(\frac{n}{1800} \right)^{-10.4924} \left(\frac{f_z}{0.05} \right)^{-4.6506} \left(\frac{a_p}{3.0} \right)^{-0.1372}, \quad (6)$$

with surrogate log-space $R^2 = 0.9800$ over 345 crossing scenarios. The high speed exponent should not be overinterpreted. It warns that the local speed range is narrow and speed is partly confounded with other load effects.

Using only uncensored NUA threshold crossings gives

$$T_{\min} = 16.78 \left(\frac{n}{1800} \right)^{-12.6655} \left(\frac{f_z}{0.05} \right)^{-4.1634} \left(\frac{a_p}{3.0} \right)^{-2.4111}, \quad (7)$$

with log RMSE 0.3784. This observed-life equation is less robust because it discards censored runs, but it independently supports a strong negative feed effect.

6.5 NASA material-aware life model

NASA is handled separately because all local NASA threshold labels cross the selected threshold and because material and initial wear carry strong effects. The material-aware AFT model includes feed, depth, material and initial wear.

The NASA AFT fit has event MAE 2.5498, RMSE 3.1544, $R^2 = 0.9689$, $\sigma = 0.2459$ and negative log likelihood 43.6327. The high R^2 is promising, but it should be interpreted with the small case count and fixed-speed design in mind.

7 Digital-twin interpretation

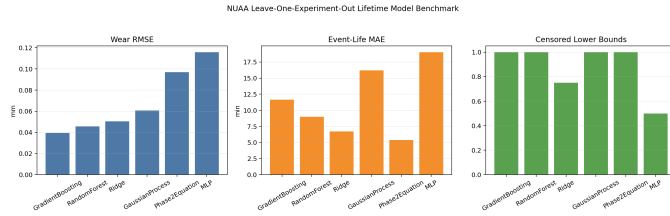
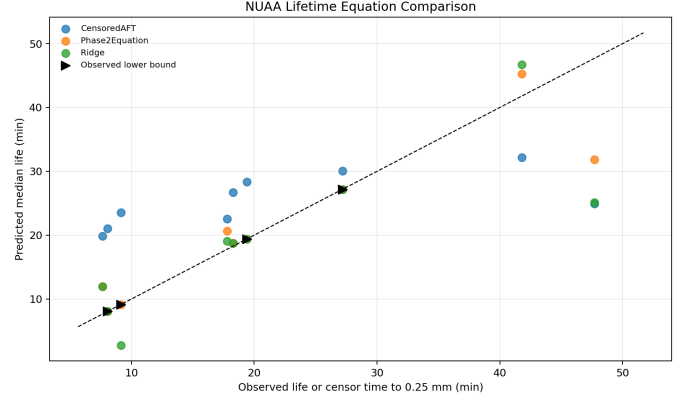
The repository includes simulator-oriented model and front-end artifacts. The simulator behavior is driven by the piecewise law and the load-intensity function. It should be understood as a research digital-twin prototype: it maps operating conditions to wear trajectories and life estimates, but its uncertainty bands and known failure modes should remain visible.

A practical decision workflow follows four steps. First, estimate the expected wear trajectory for the candidate load condition. Second, track sensor health markers, especially vibration and acoustic-emission modal features, for deviation from the expected trajectory. Third, use survival-aware lifetime outputs when threshold crossing is the primary decision. Fourth, treat high-speed exponents and short-window calibrations cautiously because they are less stable in the local data.

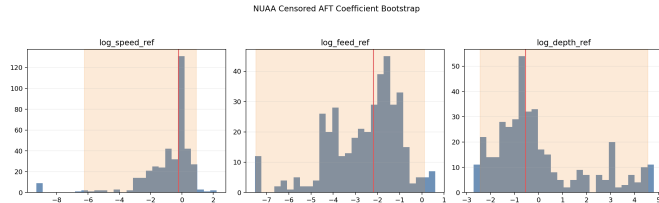
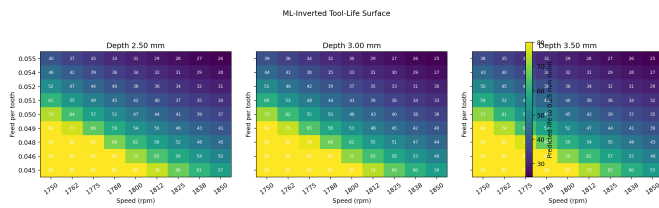
The simulator is therefore best used for scenario ranking and explanation. A low-load condition can be compared against a baseline or high-load condition to show how predicted life changes. However, the result should not be presented as a

Table 12: NUAA trained wear models and inverted event-life performance.

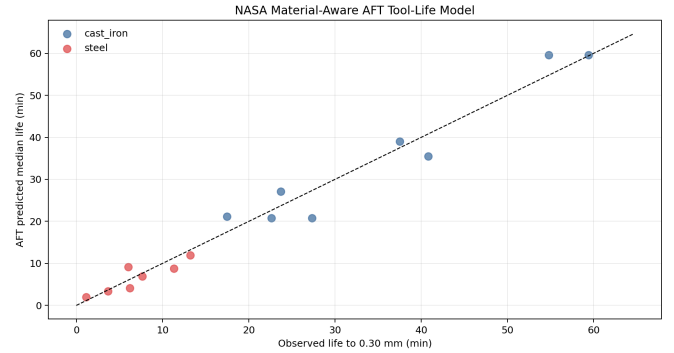
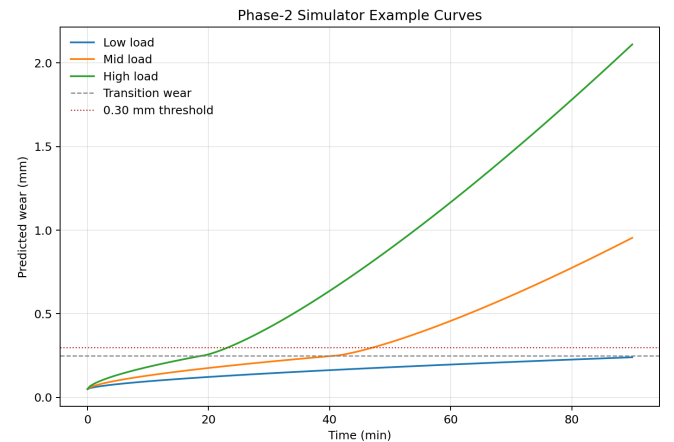
Model	Wear RMSE	Wear MAE	Wear R^2	Event-life MAE	Censor-bound rate
PiecewiseEquation	0.0969	0.0460	-0.8811	5.3832	1.00
Ridge	0.0505	0.0407	0.4883	6.7010	0.75
Random Forest	0.0457	0.0337	0.5817	8.9847	1.00
Gradient Boosting	0.0394	0.0310	0.6896	11.6555	1.00
Gaussian Process	0.0607	0.0459	0.2631	16.1923	1.00
MLP	0.1159	0.0956	-1.6890	19.0334	0.50

**(a)** Wear and life benchmark**(b)** Life equation comparison**Figure 14:** Wear-model inversion changes the model-selection problem. Smooth threshold behavior matters.**Table 13:** NUAA AFT bootstrap coefficients.

Coefficient	2.5%	Median	97.5%
Intercept	2.7199	3.2103	3.8286
log speed	-6.2471	-0.2186	0.9129
log feed	-7.4945	-2.1886	0.1153
log depth	-2.4450	-0.5467	4.5558
sigma	0.0000	0.1826	0.6821

**Figure 15:** NUAA AFT bootstrap uncertainty. Sparse events and censoring make speed and depth effects weakly identified.**Figure 16:** ML-inverted NUAA life surface. The surface is useful for scenario ranking but must be interpreted within the training design limits.**Table 14:** NASA material-aware AFT model summary.

Term	Coefficient	Mult. effect
Intercept	4.0880	59.6220
log feed reference	-0.3874	0.6788
log depth reference	-1.1356	0.3212
material: steel	-1.6069	0.2005
initial wear	-3.1201	0.0442

**Figure 17:** NASA material-aware AFT predictions. Material and initial wear are strong local drivers.**Figure 18:** Example simulated wear curves. The piecewise law produces slow early growth followed by late acceleration.

deployment-ready replacement policy until broader full-life milling datasets are ingested and the life model is validated under leave-dataset-out tests.

8 Limitations and risk

The main limitations are direct consequences of the available data. The speed exponent is unstable in bootstrap and extracted life equations because the local speed range is narrow and correlated with other design variables. NASA provides strong late-stage and sensor evidence, but its feed units, material labels and fixed-speed design make direct pooling with NUAU unsafe. PHM2010 is valuable for early-regime transfer, but the local use of PHM2010 has only three cutter groups. Gradient Boosting has the best NUAU pointwise wear RMSE but worse event-life MAE after threshold inversion. Per-run scalar calibration from short early windows overcorrects in NUAU hold-out tests. Finally, the simulator can look more decisive than the evidence supports unless intervals, limits and dataset provenance are shown.

9 Conclusions and next work

The project now contains a coherent research program for milling tool wear and tool-life forecasting. The strongest technical result is not a single machine-learning score. It is the combination of interpretable physics, signal evidence, grouped validation, threshold inversion and censoring-aware life modeling. The piecewise wear law explains why early and late data should not be forced into one undifferentiated fit. VMD and SHAP analyses show which sensor signatures carry wear information. The lifetime extraction pass shows that training a model and inverting it into life is possible, while also showing that the best wear regressor is not necessarily the best life predictor.

The practical conclusions are clear. Feed is the most reliable local load driver. Late-stage material behavior matters. Censoring must be modeled rather than ignored. Short-window scalar calibration is not yet dependable. The current digital twin is ready for research simulation, scenario ranking and explanation, but it needs larger full-life validation before industrial deployment.

The next research step is structured data expansion rather than simply adding more model classes. Each new dataset should be mapped into a common life schema: tool identifier, tool material, coating, cutter geometry, workpiece material, spindle speed or cutting speed, feed per tooth or feed rate, axial and radial depth, wear measurement type, threshold definition, native time base, sensor channels, sampling rate and event/censoring indicator. After ingestion, the piecewise law should be refit, leave-dataset-out validation should be run, scalar calibration should be replaced by a monotonic state-space update, and survival models should combine physics-law covariates with stable VMD summaries.

Table 15: Research risk matrix.

Risk	Severity	Current evidence	Recommended response
Speed exponent overinterpretation	High	Wide bootstrap interval and large ML-inverted exponent	Report speed sensitivity as provisional and prioritize new experiments with wider speed variation.
Dataset pooling mismatch	High	Different units, sensors and thresholds	Maintain dataset-specific roles and use transfer learning only after unit harmonization.
Calibration overcorrection	Medium	Calibrated RMSE worse than uncalibrated in NUAA	Replace scalar calibration with Bayesian or state-space updating.
Small event count	High	NUAA has 5 events and 4 censored records	Add full-life datasets and keep censoring explicit.
VMD feature overfitting	Medium	Many modal features relative to NASA sample count	Use grouped validation, feature stability and sparse model variants.
Simulator misuse	Medium	Curves can look decisive despite uncertainty	Surface intervals, limits and dataset provenance in the UI.

Table 16: Main equations produced by the research.

Model	Equation
Condition intensity	$\phi = (n/1800)^{0.359967} (f_z/0.05)^{3.956013} (a_p/3.0)^{0.688073}$
Early wear	$VB(t) = 0.018580943 \phi t^{0.64}$
NUAA censored AFT	$T_{\min} = 24.93(n/1800)^{-0.0187} (f_z/0.05)^{-1.7662} (a_p/3.0)^{-0.3768}$
Ridge-inverted life surface	$T_{\min} = 47.52(n/1800)^{-10.4924} (f_z/0.05)^{-4.6506} (a_p/3.0)^{-0.1372}$
Observed-life equation	$T_{\min} = 16.78(n/1800)^{-12.6655} (f_z/0.05)^{-4.1634} (a_p/3.0)^{-2.4111}$

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